

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1040

Newmark 138 kV
March 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1040. The Interconnection Customer proposes a 39.6 MW generating facility interconnecting at the Newmark 138 kV substation in the Delaware area. The request was received on August 16, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 329 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$341,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Navy Mole - Easton 2 138.0 kV	Easton 2 - Cady 345.0 kV	116.0	1,253	6.9
Easton 2 345 kV/138 kV Transformer	Cady - Easton 2 345.0 kV	116.0	1,253	6.9
Ninemile Pt Substation - Longmire 138.0 kV	Cady - Dodd Road Substation 345.0 kV	111.8	404	6.0
Las Lomas - Newmark 138.0 kV	Vienna Operations - Easton 2 345.0 kV	107.9	260	39.1
Limon - Bristol 138.0 kV line	Cady - Easton 2 345.0 kV	107.2	355	14.9
Wheatland Sta - Warren F Sam Beasley Generation Sta 138.0 kV	Dodd Road - Cady 345.0 kV	104.0	223	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Navy Mole - Easton 2 138.0 kV	Rebuild line with higher-capacity conductor	\$860,000	0% (\$1,000)
Easton 2 345-138 kV Transformer	Replace transformer with higher-rated unit	\$14,406,000	0% (\$25,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Longmire - Ninemile Pt 138.0 kV L	Rebuild line with higher-capacity conductor	\$14,188,000	0% (\$59,000)
Las Lomas - Newmark 138.0 kV	Reconductor line	\$4,346,000	4% (\$186,000)
Limon - Bristol 138.0 kV line	Reconductor line	\$1,945,000	1% (\$27,000)
Warren F Sam Beasley Generation Sta - Wheatland Station 138.0 kV	Reconductor line	\$3,766,000	1% (\$43,000)
	Total allocated to project		\$341,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.